Ji 2020 cSCC — analysis summary

RUN 4 OF 10

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

Harmony(θ=5) on sample over 20 PCs: patients thoroughly intermixed, one connected kNN component, one continuous body. A cycling lobe (35% of cells) is a proliferation confound → scoped to the developmental compartment for the trajectory.

Gate 2 — module geography → GO (non-branching)

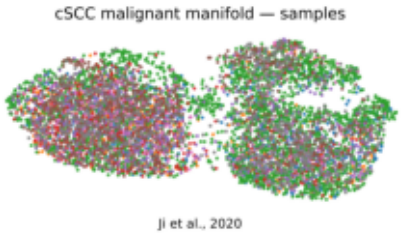
Modules present and structured, but NON-branching: Gast & pSom OVERLAP (smoothed r=+0.11) and JOINTLY oppose Crest ((Gast+pSom)–Crest r=−0.88). No Gast-arm vs pSom-arm divergence — a principal-tree Y has both terminal arms leaning the same way (Gast<pSom); Crest = inverse terminal gradient. Matches the linear Sun/Kürten references, not the branching Puram hub-and-spoke.

Gate 3 — hub cross-repression → NA

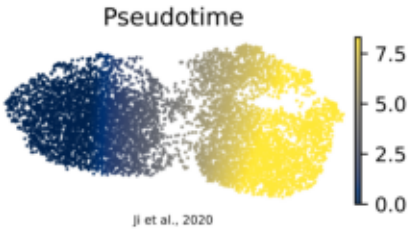
Not applicable — no bifurcation. (Context: module cross-repression on the Mixed convergence zone is r=−0.50, near the Puram anchor ≈−0.49 — the §1 'some cross-repression may persist' clause within the overlap case, not sufficient for branching without two spatially-resolved arms.)

STEP FIGURES

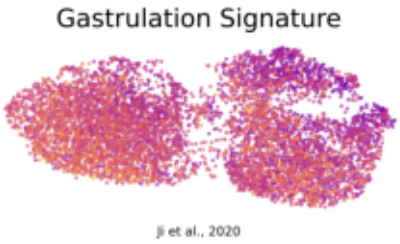
Manifold — samples



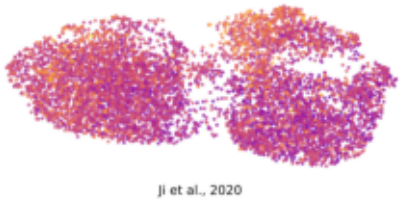
Pseudotime



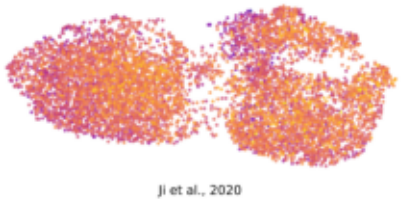
Gastrulation



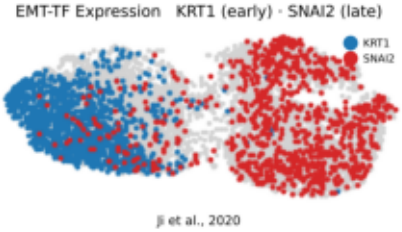
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10,598 malignant cells (8/10 samples ≥100 → 10,531; developmental compartment 6,897 after scoping the cycling lobe).

Verdict.
Embryonic programs reproduce as a NON-BRANCHING gastrulation/somitogenesis → neural-crest gradient (Gate 1 GO / Gate 2 GO non-branching / Gate 3 NA) — not the Puram-type bifurcation. Pseudotime gradient Gast ρ=−0.50, pSom ρ=−0.39, Crest ρ=+0.31. EMT/lineage TFs KRT1 (early) → SNAI2/SLUG (late).

Parameter choices

Min-cell threshold	≥100 (clean 6.7× count gap 268→40; keeps 8/10)	**Integration**	Harmony θ=5 (swept 0/2/5/10, visual mixing); PCs=20, no scaling
Cycling scope	per-Leiden cycling fraction >0.5 (3,634 cells; Gast-independent)	**Geometry**	NON-BRANCHING — principal curve (ElPiGraph), 2 tips / 0 forks, fold-free (ratio 2.97)
Root	Mixed (Gast/pSom-high, Crest-low origin — a tip, not the Crest-high terminus)	**Annotation**	data markers (KRT1/10, MMP10/PTHLH/LAMC2, TP63) + Ji 2020 subtypes (TSK, Tumor-KC)
EMT-TFs	KRT1 (early) vs SNAI2/SLUG (late)		